



2669 Martin Luther King Jr Blvd
Youngstown, OH 44510

VALLOUREC STAR LP
Seamless Tubular Products

Certified Test Report
06/17/2025
Revision: 0

Order: Y-221129443-001

Desc: Seamless Line Pipe- Normalized Rolled

Customer: MRC Global (US) Inc.

SLN / (Heat): PG0098

Grade: API 5L X42/B N/ BN PSL2 46th Edition May 01, 2019

Customer Order: S 9W3 684173

Split: 00

OD(In): 4.500

Wall(In): 0.337

Lbs/Ft: 15.00

Length: Double Random

End Finish: Plain End Beveled

Comments: Melted and Manufactured in Youngstown, Ohio USA

Pipe is also manufactured to: ASTM A106B/C-19a, ASTM A53B-24, ASME SA106B/C-2023, ASME SA53B-2023. Vallourec Star's QA program meets the requirements of Directive 2014/68/EU and DIN EN10204-2004 Type 3.1.

Satisfies NACE MR0175/ISO15156:2015 and ANSI/NACE MR0103/ISO17945:2015. Material has been produced using silicon killed carbon steel with aluminum fine grain practice.

No repair welding has been conducted. Material is free from Mercury.

S109-D MRC

Mechanical Properties

Orientation	Type	Tensile Size			Strength (ksi)			Y/T Ratio	Elongation		
		Width (in)	Thick (in)	Area (sq in)	Yield	Tensile			Gage Length	% Elong	% RA
Longitudinal	Strip	0.990	0.370	0.3700	47.2	72.6	0.65		2	38.9	58.7
Required Min/Max Yield Strength (ksi): 42.1/71.8					Required Min/Max Tensile Strength (ksi): 70/95			Calculated Max Internal Yield (psi): 6187			

Required Min Elongation(%): 30

Chemical Analysis

Chem Spec: 61M

WT%	C	Mn	P	S	Si	Cu	Ni	Cr	Mo	Sn	Nb	V	Al	Ca	B	Ti	N	C.E.
Heat	0.20	0.68	0.010	0.004	0.24	0.23	0.07	0.09	0.02	0.011	0.000	0.001	0.029	0.0018	0.0003	0.002	0.0070	0.35
Product 1	0.20	0.68	0.010	0.003	0.24	0.23	0.07	0.09	0.02	0.009	0.000	0.001	0.031	0.0018	0.0001	0.002	0.0072	0.36
Product 2	0.21	0.67	0.010	0.003	0.24	0.23	0.07	0.09	0.02	0.008	0.000	0.001	0.031	0.0020	0.0001	0.002	0.0066	0.36

Charpy Impact Testing - Not required per API.

This material has been produced and tested in accordance with the requirements of applicable specifications unless otherwise listed below. We hereby certify that the above test results are representative of those contained in the records of the company. Any modification to this certification as provided by Vallourec Star without the expressed written consent of Vallourec Star negates the validity of this test report. Vallourec Star is not responsible for the inability of this material to meet specific applications.

THIS CERTIFICATE IS NOTARIZED ONLY WHEN REQUESTED.

Signed: *Jeff Schultz*

Name: Jeff Schultz, Lab Supervisor

Date Approved: 06/17/2025

This document is uncontrolled when printed. Always refer to the MTR System for the most current version. FS-VMS-8.2.4.1.1 Rev 7

Supplemental Requirements

Flattening Test	Passed
Hardness 3 Point	76.5 avg HRBW
Hydrostatic Pressure	4730 psi
Hydrostatic Hold Time	5 seconds



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Inspection Company: L Bay

Non-Destructive Evaluation Criteria

UT CHANNELS

ID

10% Longitudinal

10% Transverse

OD

5% Longitudinal

5% Transverse

WALL THICKNESS CHECK

87.5% Wall Reduction

Required Volumetric Coverage: 50%

ADDITIONAL REQUIREMENTS

Notch Length $\leq 1.00"$

Notch Width .020" Max

Untested Ends Cropped

OD

Visual Insp. Full Length - White Light

ID

Visual Insp. Pipe Ends - White Light